Fiscal Year:	FY 2019	**Task Last Updated:**	FY 07/18/2019
PI Name:	Lee, Stuart M.C. Ph.D.		
Project Title:	Metabolomic and Genomic Markers of Atherosclerosis as Related to Oxidative Stress, Inflammation, and Vascular Function in Twin Astronauts		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	281-483-3726
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City:	Houston	**State:**	TX
Zip Code:	77058-2749	**Congressional District:**	36
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-TWINS
Start Date:	07/15/2014	**End Date:**	12/31/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Charles, John	**Contact Phone:**	281-483-7224
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Grant number change to 80NSSC18K0037 and end date change to 12/31/2018 per NSSC information (Ed., 1/9/19) NOTE: Extended to 9/30/2018 per PI (Ed., 5/7/18) NOTE: Extended to 4/18/2018 per NSSC information (Ed., 8/25/17)		
Key Personnel Changes/Previous PI:	No changes		
COI Name (Institution):	Hargens, Alan Ph.D. (University of California, San Diego) Macias, Brandon Ph.D. (University of California, San Diego) Rana, Brinda Ph.D. (University of California, San Diego) Sears, Dorothy Ph.D. (University of California, San Diego) Smith, Scott Ph.D. (NASA Johnson Space Center) Stenger, Michael Ph.D. (Wyle Laboratories, Inc.)		
Grant/Contract No.:	80NSSC18K0037 ; NNX14AN75G		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	Astronauts participating in long-duration missions may be at an increased risk of oxidative stress and inflammatory damage due to radiation, psychological stress, altered physical activity, nutritional insufficiency, and hyperoxia during extravehicular activity. This proposal extends a current NASA-funded investigation to determine whether these spaceflight factors contribute to an accelerated progression of atherosclerosis. We propose to measure blood and urine biomarkers of oxidative stress and inflammation as well as arterial structure and function (carotid intima-medial thickness and brachial artery flow-mediated dilation) in one twin astronaut before, during, and after long-duration spaceflight and in his twin serving as a ground-based control. Furthermore, we will measure metabolomic and genomic (DNA methylation, mRNA gene expression, telomere length) markers to elucidate the molecular mechanisms involved. We predict that the space-flown twin will experience elevated biomarkers of oxidative stress and inflammatory damage, altered arterial structure and function, accelerated telomere shortening, dysregulation of genes associated with oxidative stress and inflammation, and a metabolic profile shift that is associated with elevated atherosclerosis risk factors. Conversely, these will not be observed in the ground-based twin. This study of twins affords a unique opportunity to examine the spaceflight-related atherosclerosis risk independent of the confounding factors of different genotypes.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work seeks to improve understand of the potential cardiovascular disease risk among long-duration space flown astronauts and biomarkers of atherosclerotic risk in normal Earth-bound populations.
Task Progress:	The study has been completed, and the results have been published in a peer-reviewed scientific journal (Science); see Bibliography section. Our primary findings include a thickening of the carotid intimia-media layer during and after spaceflight and increased levels of biomarkers of inflammation and oxidative stress during the mission. These changes occurred early in the mission and importantly did not increase later in the flight.
Bibliography Type:	Description: (Last Updated: 11/18/2025)
Articles in Peer-reviewed Journals	Garrett-Bakelman FE, Darshi M, Green SJ, Gur RC, Lin L, Macias BR, McKenna MJ, Meydan C, Mishra T, Nasrini J, Piening BD, Rizzardi LF, Sharma K, Siamwala JH, Taylor L, Vitaterna MH, Afkarian M, Afshinnekoo E, Ahadi S, Ambati A, Arya M, Bezdan D, Callahan CM, Chen S, Choi AMK, Chlipala GE, Contrepois K, Covington M, Crucian BE, De Vivo I, Dinges DF, Ebert DJ, Feinberg JI, Gandara JA, George KA, Goutsias J, Grills GS, Hargens AR, Heer M, Hillary RP, Hoofnagle AN, Hook VYH, Jenkinson G, Jiang P, Keshavarzian A, Laurie SS, Lee-McMullen B, Lumpkins SB, MacKay M, Maienschein-Cline MG, Melnick AM, Moore TM, Nakahira K, Patel HH, Pietrzyk R, Rao V, Saito R, Salins DN, Schilling JM, Sears DD, Sheridan CK, Stenger MB, Tryggvadottir R, Urban AE, Vaisar T, Van Espen B, Zhang J, Ziegler MG, Zwart SR, Charles JB, Kundrot CE, Scott GBI, Bailey SM, Basner M, Feinberg AP, Lee SMC, Mason CE, Mignot E, Rana BK, Smith SM, Snyder MP, Turek FW. "The NASA Twins Study: A multidimensional analysis of a year-long human spaceflight." Science. 2019 Apr 12;364(6436):eaau8650. https://science.sciencemag.org/content/364/6436/eaau8650.long ; PubMed PMID: 30975860 , Apr-2019